Sukhrob Ilyosbekov

Senior Software Engineer · AI/ML & Computer Vision

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Summary

Machine-learning engineer with published computer-vision research and 9+ years building production software. I take deep-learning and LLM work past the prototype stage: imaging models pulled into hospital workflows, an AI freight dispatcher, and multi-model pipelines running in live SaaS. The full-stack background is what gets the models deployed and keeps them running once they're out.

Technical Skills

AI/ML: PyTorch, TensorFlow, EfficientNet, YOLO, GradCAM++, OpenCV, computer vision, LLM/agentic systems, MCP, Claude/GPT/DeepSeek APIs

Languages: Python, C#, C++, TypeScript, JavaScript, Kotlin

ML Tooling & Data: FastAPI, OHIF/DICOM, PostgreSQL, MongoDB, Redis, Prisma, NumPy/pandas

Backend: ASP.NET Core, EF Core, SignalR, Node.js, Bun, NestJS, ElysiaJS

Frontend: React, Next.js, Angular, Blazor, Tailwind CSS

Cloud & DevOps: AWS (S3, Lambda, Cognito, Amplify), Azure, Docker, Kubernetes, GitHub Actions

Education

Northeastern University, M.S. in Computer Science, Boston, MA — *Jan 2024 – May 2026*

Suffolk University, B.S. in Computer Science, Boston, MA — *Sep 2021 – May 2023*

Professional Experience

Senior AI/ML Engineer | EmTech Care Labs | Portland, ME — *Jan 2025 – Present*

- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform, including a risk model that flags patients showing early signs of decline from daily activity and care-log data so caregivers can step in sooner.
- Added an NLP pipeline that reads free-text clinical notes and pulls out structured details (symptoms, medications, care events), cutting the manual charting load for the care team.
- Stood up the ML infrastructure under HIPAA constraints, with de-identified training data, model serving behind the API, and monitoring. The platform cleared an outside HIPAA readiness review with no critical findings.

Senior Machine Learning Engineer | AllFactors | Santa Clara, CA — *Oct 2021 – Dec 2024*

- Built property-valuation and market-forecasting models on real-estate data that powered the product's analytics, turning raw pricing and inventory feeds into predicted values and trend signals.
- Designed the real-time feature pipeline (SignalR + SQL Server change tracking) that fed live market data to the models and pushed updated predictions to hundreds of concurrent users with sub-second latency.
- Productionized the models into the customer-facing dashboards and added automated tests around the data and prediction flows.

Software Engineer | Smart Meal Service | Russia — *Sep 2020 – Jul 2021*

- Built the software for a self-service food kiosk and robotic cashier in C# / .NET, integrating with payment terminals and hardware over low-level protocols, plus the customer ordering app and an admin dashboard.
- Added automated order routing that sent each order straight to the right kitchen station, and a hardware abstraction layer so one codebase ran across three machine models.

Game Developer | Pentalight Technology | Malaysia — *Mar 2020 – Feb 2021*

- Built the multiplayer networking for a VR smart-city simulation in Unity, syncing every participant's view in real time, plus spatial audio, hand-tracking, and a desktop spectator client.

Earlier: Freelance Developer at Freelancer.com (2019 – 2020), building Python NLP and .NET applications. Game Developer at EC Dev Team, Uzbekistan (2016 – 2019), led development of an RTS mod for Paradox's Clausewitz Engine.

Selected AI/ML Projects

MorphoCLIP GitHub | Paper

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Built a text-supervised contrastive model, pairing frozen vision/language backbones (DINOv3, BioClinical ModernBERT) with trainable projection heads, that matches Cell Painting microscopy images of drug- and gene-perturbed cells to natural-language treatment descriptions, evaluated on the CPJUMP1 benchmark (51 plates, 3M+ images). Published on arXiv with two Northeastern University co-authors.

Localize, Don't Beautify https://arxiv.org/abs/2608.02841

Image-editing APIs, inpainting, ArcFace, prompt engineering

- Studied why commercial image-editing APIs over-edit the face during cosmetic-surgery preview generation and how much of that can be fixed client-side without touching the model; tested prompt-only, masked-compositing, and model-based inpainting across six editors, and found simple masking beat model-based inpainting on localization accuracy. Published on arXiv.

MelanomaNet GitHub | Paper

PyTorch, EfficientNet V2, GradCAM++, OpenCV

- Trained a deep-learning model to classify skin lesions across nine diagnostic categories on a standard public dermatology dataset of 25K+ images, reaching 0.86 weighted F1.
- The research contribution is interpretability: the model produces heatmaps of what it looked at, decomposed along the same ABCDE criteria (Asymmetry, Border, Color, Diameter, Evolution) dermatologists already use, so a clinician can follow and trust the reasoning rather than getting an opaque score. Published on arXiv.

Med Image Scanner https://github.com/suxrobgm/med-image-scanner

FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- HIPAA-compliant platform that pulls X-ray, CT, and MRI scans directly from hospital imaging systems and runs PyTorch detection models on them, flagging pneumonia on chest X-rays and intracranial hemorrhage on head CTs.
- Renders model predictions as overlays inside a real radiology viewer (OHIF), and handles patient data safely throughout, with on-the-fly de-identification, audit logging, and role-based access.

Bookshelf Scanner https://github.com/suxrobgm/bookshelf-scanner

YOLO, Moondream2 VLM, FastAPI, Angular

- Two-stage computer-vision pipeline: a YOLO segmentation model isolates each individual book spine from a shelf photo, then a vision-language model reads the title and author off each spine.
- Wrapped it in a web app for uploading photos, correcting misreads, and exporting the recognized library to Goodreads or LibraryThing.

LogisticsX https://logisticsx.app | GitHub

Claude API, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin KMP, PostgreSQL

- Built an AI dispatcher that matches freight loads to trucks, checks federal driver hours-of-service rules, and plans multi-stop routes on its own, turning a manual 15-minute decision into a near-instant one. Runs on the Claude API through a custom tool-use agent with its own set of tools.
- Shipped it as a real product: web portals for dispatchers, customers, and brokers, a cross-platform driver app, and a multi-tenant .NET backend integrated with the major freight load boards and tracking providers.

OGStack https://ogstack.dev | GitHub

Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe

- Built a multi-model AI layer that routes work across Claude, GPT, and DeepSeek to read a web page and match its branding, then generates an on-brand social-share image, including AI-generated backgrounds.
- Productionized it as a SaaS serving cached images from a global CDN; it generated 10K+ images during the beta.